MIVA OPEN UNIVERSITY

STUDY/COURSE GUIDE

COS 203: Discrete Structures



Course Material Development Team

Course Developer/Writer: Dr Morolake Lawrence, PhD, Computer Science.

Course Editor: Obinna Awoke, MSc, Computer Science.

Course Coordinator: Dr Oluwasegun Adelaiye, PhD Computer Science

Course Information

Course Code: COS 203
Course Title: Discrete Structures
Credit Unit: 2
Course Status: C
Learning Hours (LH): 30

Ice Breaker

Discrete structures are the building blocks of modern computer science, and understanding them is key to developing robust algorithms and logical systems. Whether you're working on database design, cryptography, or network theory, this course will give you the tools to think logically and mathematically. Let's dive into how discrete mathematics shapes the digital world and empowers you to solve complex problems efficiently.

Contents

Ice Breaker	2
Contents	3
Introduction	4
Course Aims	4
Course Objectives	4
Working Through This Course	5
What You Will Learn in the Course	5
Course Materials	6
Study Sessions	7
Set of Textbooks/References	7
Assignment File	8
Glossary	8
Presentation Schedule	9
Course Marking Scheme	9
Course Overview	10
How to Get the Best from the Course	11
Assessment	11
Tutor-Marked Assignments (TMAs)	12
Final Examination and Grading	12
Tutors and Tutorials	12
Summary	**13**

Introduction

COS 203: Discrete Structures covers essential mathematical concepts that underpin computer science and technology. The course introduces key topics like propositional and predicate logic, proof techniques, combinatorics, and recurrence relations. By the end of this course, you will have developed strong problem-solving skills and be able to apply logic and counting techniques to real-world applications.

Course Aims

This course is designed to:

- Teach the fundamental principles of discrete structures in computer science.
- Provide a strong foundation in logic, set theory, proof techniques, and combinatorics.
- Equip you with the tools to solve problems using counting methods, permutations, combinations, and recurrence relations.
- Enable you to construct sound logical arguments and apply mathematical principles to computer science problems.

Course Objectives

By the end of this course, students will be able to:

1. Convert logical statements from informal language to propositional and predicate logic expressions.

2. Describe the strengths and limitations of propositional and predicate logic, outlining basic proof techniques.

3. Apply proof techniques correctly in constructing sound arguments and use the pigeonhole principle in formal proofs.

4. Compute permutations and combinations, interpret their meaning, and solve problems in context.

5. Map real-world problems to appropriate counting formalisms and solve recurrence relations.

Working Through This Course

There are twelve study units in this course, each corresponding to one week of study. The content is divided into logical segments with exercises to reinforce learning. Be sure to read each unit thoroughly and complete all exercises. Engage with the assignments as they are part of the continuous assessment and contribute to your final grade.

What You Will Learn in the Course

In **COS 203**, you will:

- **Propositional Logic:** Learn the syntax and semantics of propositional logic and how to represent logical statements in a formal language.

- **Predicate Logic:** Understand how predicate logic extends propositional logic to include quantifiers, predicates, and logical expressions involving variables.

- **Sets and Functions:** Study basic set theory and the operations on sets, as well as functions and their properties.

- **Proof Techniques:** Learn techniques like mathematical induction, direct proof, and contradiction to construct rigorous logical proofs.

- **Combinatorics:** Master the fundamentals of counting, including permutations and combinations, and apply these methods to solve practical problems.

- **Recurrence Relations:** Learn to solve and apply recurrence relations in problems related to algorithms and computer science.

- **Discrete Probability:** Understand the concepts of discrete probability, including events, outcomes, and the laws governing probability.

Course Materials

1. **Course Guide**
2. **Study Units**
3. **Recommended Textbooks and Resources**
4. **Assignments**
5. **Presentation Schedule**

Study Sessions

This course consists of 12 study units, each corresponding to one week of study. Topics covered in each session include:

- **Study Session 1:** Propositional Logic
- **Study Session 2:** Predicate Logic
- **Study Session 3:** Sets and Functions
- **Study Session 4:** Sequences and Summation
- **Study Session 5:** Proof Techniques
- **Study Session 6:** Mathematical Induction
- **Study Session 7:** Inclusion-Exclusion and Pigeonhole Principle
- **Study Session 8:** Permutations and Combinations (with and without repetitions)

- **Study Session 9:** The Binomial Theorem
- **Study Session 10:** Discrete Probability
- **Study Session 11:** Recurrence Relations
- **Study Session 12:** Applications of Recurrence Relations

Set of Textbooks/References

1. **"Discrete Mathematics and Its Applications" by Kenneth H. Rosen**
    - **Relevance:** A comprehensive text on discrete mathematics, covering logic, set theory, and combinatorics.
2. **"Mathematics for Computer Science" by Eric Lehman, F. Thomson Leighton, and Albert R. Meyer**
    - **Relevance:** A textbook focused on discrete structures, logic, and proof techniques for computer science students.
3. **"Discrete Mathematics with Applications" by Susanna S. Epp**
    - **Relevance:** Provides a solid foundation in discrete mathematics with an emphasis on problem-solving techniques.

Assignment File

The assignment file contains detailed instructions for completing and submitting your assignments. Be sure to read the instructions carefully and follow them as directed.

Glossary

- **Propositional Logic:** A branch of logic that deals with propositions and their connectives, using truth values (true/false).
- **Predicate Logic:** An extension of propositional logic that includes quantifiers and predicates to handle statements involving variables.
- **Set Theory:** A branch of mathematics that studies sets, which are collections of objects.
- **Combinatorics:** The branch of mathematics dealing with counting, arrangement, and combination of objects.
- **Permutations:** The arrangement of objects in a specific order.
- **Combinations:** The selection of objects without regard to the order.
- **Recurrence Relation:** A mathematical relation that defines a sequence of values based on previous values in the sequence.
- **Mathematical Induction:** A method of mathematical proof typically used to prove statements about integers.
- **Pigeonhole Principle:** A principle that asserts if more objects are put into fewer containers, at least one container must contain more than one object.

- **Binomial Theorem:** A formula used to expand expressions that are raised to a power, such as (a+b)n(a + b)^n(a+b)n.

Presentation Schedule

The presentation schedule will provide the dates for submitting assignments, attending tutorials, and the final exam. Please make sure you adhere to these dates to stay on track with the course.

Course Marking Scheme

Assessment	Marks
Continuous Assessment (CA)	40%
CA1 (Multiple Choice/Essay)	20%
CA2 (Multiple Choice)	20%
Final Examination	60%
Total	**100%**

Course Overview

Units	Title of Work	Week's Activity	Assessment (end of unit)
1	Propositional Logic	1	Assignment 1
2	Predicate Logic	2	Assignment 2
3	Sets and Functions	3	Assignment 3
4	Sequences and Summation	4	Assignment 4
5	Proof Techniques	5	Assignment 5
6	Mathematical Induction	6	Assignment 6
7	Inclusion-Exclusion and Pigeonhole Principle	7	Assignment 7
8	Permutations and Combinations (with and without repetitions)	8	Assignment 8

9	The Binomial Theorem	9	Assignment 9
10	Discrete Probability	10	Assignment 10
11	Recurrence Relations	11	Assignment 11
12	Final Project and Review	12	Assignment 12

How to Get the Best from the Course

- Keep up with the weekly study units, and be sure to engage with the exercises.
- Work on assignments as soon as they are given to reinforce what you have learned.
- Seek help from your tutors or classmates if you encounter difficulties.
- Stay organised by keeping track of deadlines and presentation schedules.

Assessment

Your assessment for the course will be based on:

1. **Tutor-Marked Assignments (TMAs)**

2. **Computer-Marked Assessment (CMA)**

3. **Final Examination**

The final exam accounts for 60% of your total grade, while the continuous assessment makes up the remaining 40%.

Tutor-Marked Assignments (TMAs)

You will complete tutor-marked assignments that will test your understanding of the material. These will contribute 40% of your final grade. Be sure to submit them on time and follow the guidelines provided in the assignment file.

Final Examination and Grading

The final exam will test your theoretical understanding and ability to apply the concepts learned in the course. It will consist of multiple-choice questions, short answer questions, and problem-solving exercises. The final exam will account for 60% of your total grade.

Tutors and Tutorials

The course includes 15 hours of tutorials, which will be delivered through a combination of recorded videos (asynchronous learning), live sessions (synchronous learning), office hours, group work, and projects. Your tutorial group will be communicated to you. Feel free to relate with your tutorial facilitator who will mark, correct your assignments, and monitor your attendance/performance in tutorial-marked assignments. You should always contact your tutorial facilitator by e-mail if you have any problems with the contents of any of the study units.

Summary

In **COS 203: Discrete Structures**, you will gain essential skills in logical reasoning, combinatorics, and recurrence relations. These concepts are critical in fields like algorithm design, cryptography, and network theory. You will learn to translate real-world problems into mathematical models, use proof techniques to verify correctness, and apply combinatorics to solve problems in practical applications. This course is essential for anyone pursuing a career in computer science or related fields.